Artificial Intelligence (AI)

What is Generative AI?

- A type of computer program that creates content [based on patterns that it has learned]
  - Text, images, videos, audio
  - Examples:
    - Answer questions, rewriting text (more formal, friendly), summarising text, translating text, designing greeting cards, writing stories, creating itineraries, fixing pictures, recommender systems etc.
- Think of it as a smart assistant that “learns” from many examples, and then mixes that knowledge to produce something new

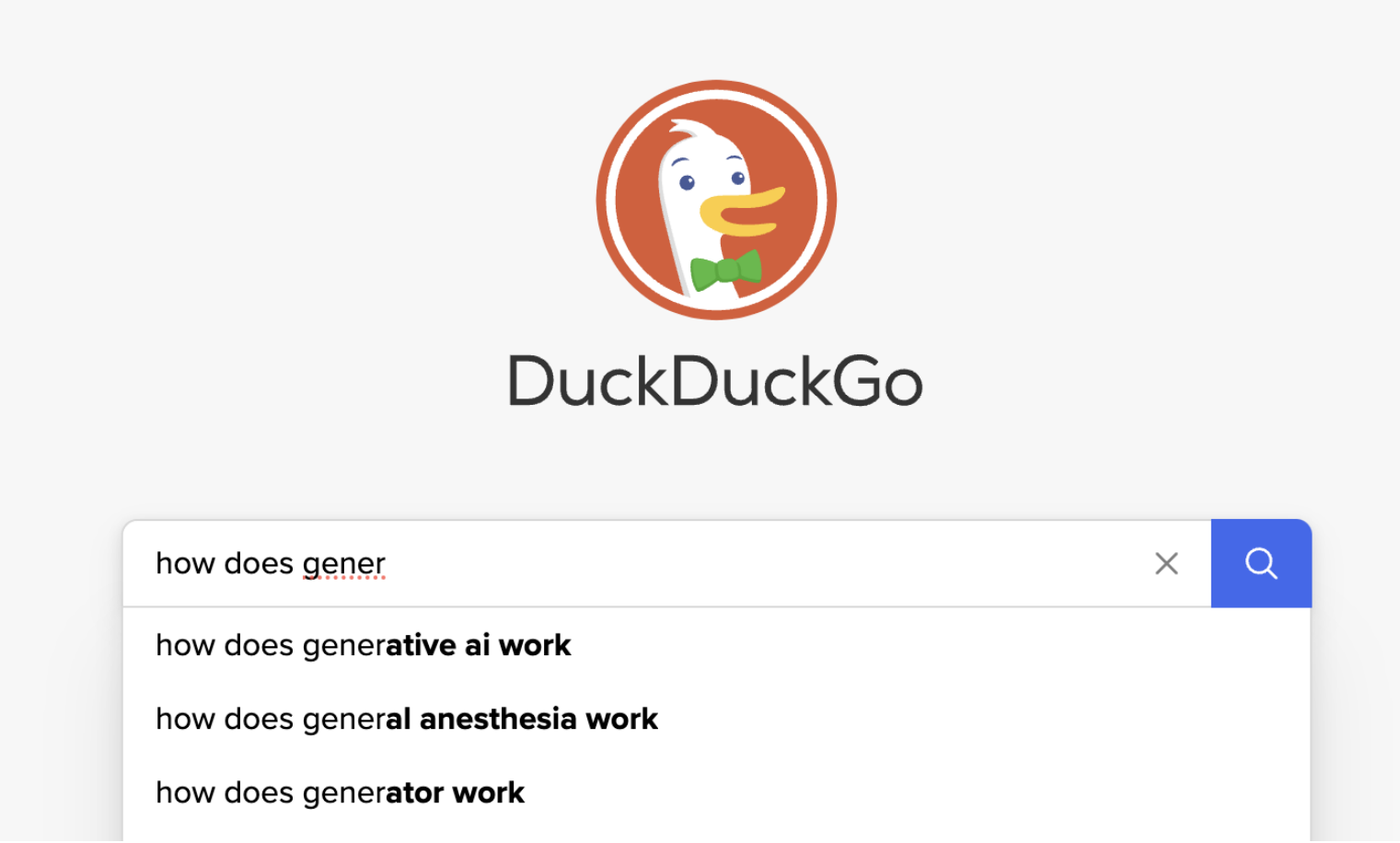
Generative AI providers

- OpenAI (ChatGPT)
- Google (Gemini)
- Anthropic (Claude)
- Mistral (Le Chat)
- Meta (Meta AI)
- Grok (X/Twitter)

Open Source / Self-Hosted:

- Meta (Llama)
- Deepseek
- Alibaba (Qwen)
- Mistral (Pixtral)

How does it work?

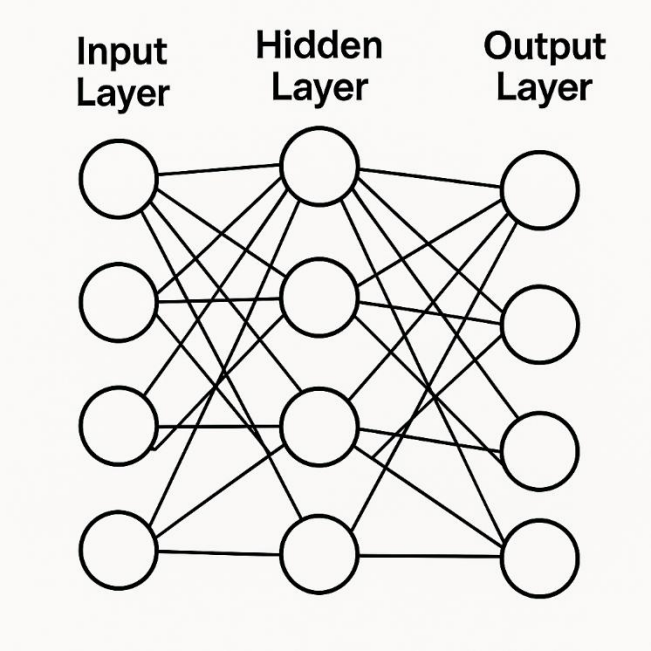


How does it *actually* work?

- The AI learn from <u>lots</u> of data (like reading many books) and then use patterns they learned to “guess” what should come next in a sentence

- Imagine learning a dance by watching billions of dance performances… then using those moves to create your own routine

- Training:
  - Internet content (websites, social media)
  - Scientific books
  - Humans help with the training by providing feedback
  - Common for grounding to happen with internet access (search



Pros of Text Generation

- Easy access to information
- Help with day-to-day tasks
  - Brainstorming, writing, summarising, planning, fixing toilet
- Entertainment

You

Can you please create a monthly meals schedule based on the meals below. Please make meals diverse (e.g., salmon dishes across weeks) but try to combine meals that share basic ingredients like courgettes, peppers, and baby tomatoes. Every Friday should be pizza. Try and have as close to an equal distribution as possible between [James] and [Rebecca] dishes every week. [Weekend] means should only be on Saturdays or Sundays, and [New] meals should always be on Saturdays. Output schedule as a table.

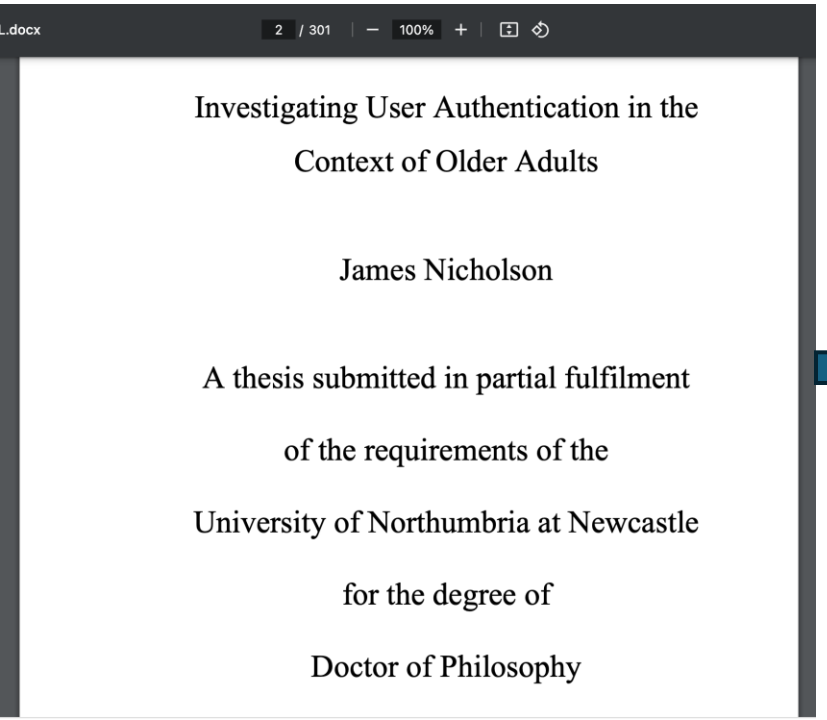
- Halloumi wraps [Rebecca]
  - Halloumi
  - Wraps

GPT 4.1 Mini

Certainly! Below is a **monthly meal schedule** laid out in a table format.

- Each week alternates roughly equal dishes between **[James]** and **[Rebecca]**.
- **Fridays** are always pizza-based (Frozen pizza for James or Pitta pizzas for Rebecca).
- **Weekend** meals are only on Saturdays or Sundays.
- **New** meals are always served on Saturdays.
- Dishes sharing core ingredients like courgettes, peppers, and baby tomatoes are grouped within the same week or close proximity where possible for efficient shopping and variety.
- Salmon dishes are spread across weeks and varied.

Week / Day	Monday	Tuesday	Wednesday
Week 1	Stir fry noodles [Rebecca]	Baked potatoes [James]	Halloumi wraps [Rebecca]
	(peppers, tenderstem broccoli)	(potatoes, tuna, cheese)	(halloumi, peppers, chili jam
Week 2	Rice bowl [Rebecca]	Fishcakes [James]	Shakshuka [Rebecca]



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Investigating User Authentication in the Context of Older Adults

James Nicholson

A thesis submitted in partial fulfilment of the requirements of the University of Northumbria at Newcastle for the degree of Doctor of Philosophy

You

Can you summarise the attached document into two paragraphs, focusing specifically on the novelty of the work

PDF document (1).pdf 6.06MB

GPT-4o Mini

The thesis by James Nicholson investigates user authentication systems, particularly focusing on older adults' performance compared to younger adults. The research presents empirical evidence that older adults face significant challenges with traditional knowledge-based authentication methods, such as Personal Identification Numbers (PINs), due to cognitive decline. Notably, the study evaluates various Graphical Authentication Systems (GAS), including Tiles, Faces, and Pictures, aiming to create more inclusive authentication methods. While traditional PINs were found to disadvantage older adults, the research revealed that graphical systems could potentially improve their performance, especially when designed with considerations for age-related memory advantages, such as the use of familiar or age-appropriate images.

A significant contribution of this work is the introduction of face-based systems for authentication that leverage the own-age effect, which suggests that older adults perform better with familiar faces. The study compares younger and older participants across different systems and time intervals, showcasing how design choices—such as eliminating image overlap and using old faces—can enhance memorability and reduce performance decrements for older users. This research not only highlights the necessity for more inclusive authentication systems but also provides a methodological framework for evaluating these systems in a way that accounts for age-related cognitive differences, thereby addressing a critical gap in existing literature on user authentication.

Cons of Text Generation

- IP theft
- Environmental costs
- Loss of critical thinking
- Job losses
- Hallucinations
- “yes man” responses
- Mental health failures

Cons of Text Generation

is lady gaga two days older th

Gemini 2.0 Flash Entire Web

are sharks older than the moon

It is likely that **sharks are older than the moon**. Sharks appeared on Earth approximately 450 million years ago, while the Moon is estimated to have formed around 4.51 billion years ago .

- **Age Difference:** Lady Gaga was born two days before Ariana Grande.

Security risks of text-based AI

- Can help scammers scale attacks (easier to carry out)
  - Phishing
  - Romance scams
  - Website generation / technical tasks

- But also:
  - Can provide information and help (as models become more accurate)
  - Can help identify phishing messages
  - Can bait scammers (and bots)

Please a selfie picture of a good looking, friendly, white man in his early 60s in a busy high street. Low res disposable camera, average rating

Perplexity Images 1 task

This is the inside of my toilet - where is the flapper?

Perplexity Sources 1 1 task ↗

The toilet in your image does not have a traditional rubber flapper, as it appears to use a modern dual-flush mechanism. In such systems, the flapper is replaced by a flush valve assembly, which is typically integrated into the blue and white components visible in your photo.

Here's how you can troubleshoot this type of toilet:

1. **Flush Valve (Blue Component)**
    - The blue section likely controls the release of water when the toilet is flushed. If water keeps running, the flush valve seal (located at the base of this assembly) may be dirty, misaligned, or worn out.
    - **Solution**: Turn off the water supply, remove the flush valve assembly by twisting or unclipping it, and inspect the seal at its base. Clean it or replace it if necessary.
2. **Fill Valve (White Component on the Right)**

Pros of Multimedia Generation

- Easy access to generating art
- Help with design process (e.g., prototyping)
- Can fix photos
- Entertainment

Cons of Multimedia Generation

- IP theft
- Proliferation of rubbish content
- Environmental costs
- Destroys artist jobs
- What is real???

Is this picture real?

- Look out for weird fingers.
- Some window weirdness.
- Also, paintbrush is backwards (inconclusive by itself).
    - Same with the palette – doesn't match the artwork, but not conclusive.

Is this picture real?

- No… but very hard (impossible?) to tell…

Image Gallery

- If you know the words, you can create the visuals
- https://www.midjourney.com/explore?tab=top_month

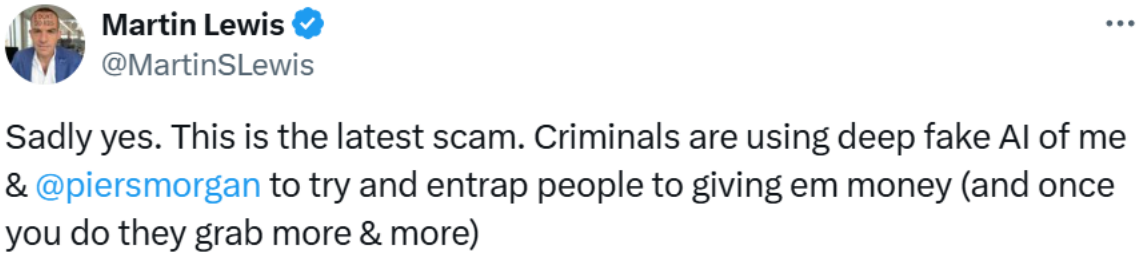
Videos can also be generated using AI

- https://www.which.co.uk/news/article/quiz-could-you-spot-a-deepfake-a5Bk99o6oaPW

Security risks of Multimodal AI

- Can help scammers be more convincing
  - Romance scams (pictures/videos)
  - Fake products (pictures)
  - Voice cloning for phone calls
  - Celebrity endorsements

- But also:
  - Can provide information and help (as models become more accurate)
  - Can help identify phishing messages
  - Can bait scammers (and bots)



Privacy Risks of Generative AI

- **Training**: Many AI systems will log user interactions to improve the models (”training”).
  - Any personal or sensitive information could be stored and accessed by third parties, or used to generate content for other people

- **Prompt Injections**: It is possible to reveal sensitive data that has been learned by the AI if you craft queries carefully (hack it)

- **Breaches**: It may be possible for personal/sensitive data that has been inputted to be identified if the Generative AI provider suffers a data breach

Key Takeaways

- Generative AI has become much more powerful in the last few years
- It is **very useful** for all kinds of things, but has **personal, environmental and societal costs** that are still emerging
- It is a good idea to try using it, at least so that you can recognise it and be aware of current capabilities
  - Remember that images and videos on the internet are now very easily faked
- But
  - Beware of “hallucinations” (made up information)
  - Don’t become dependent
  - Remember the environmental cost